Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.001 with the Adam optimizer. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the training loss.

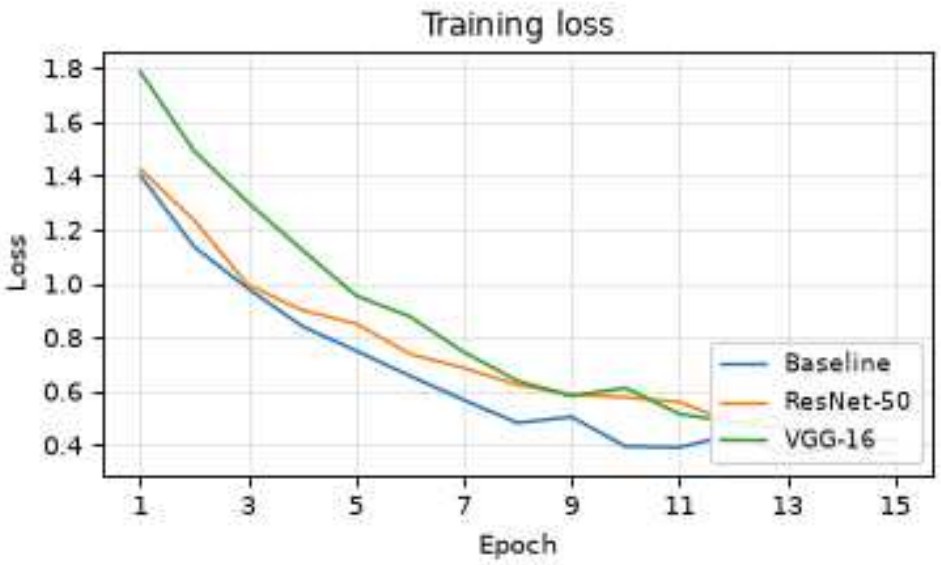


Figure 1: Training loss for the models in this experiment.

At epoch 12, the loss of ResNet-50 is 0.402. As shown in Figure 1, at epoch 9, the loss of ResNet-50 is about 0.6. ResNet-50 has a loss of 0.529 at epoch 11 (Figure 1).

Figure 2 shows the test accuracy during training.

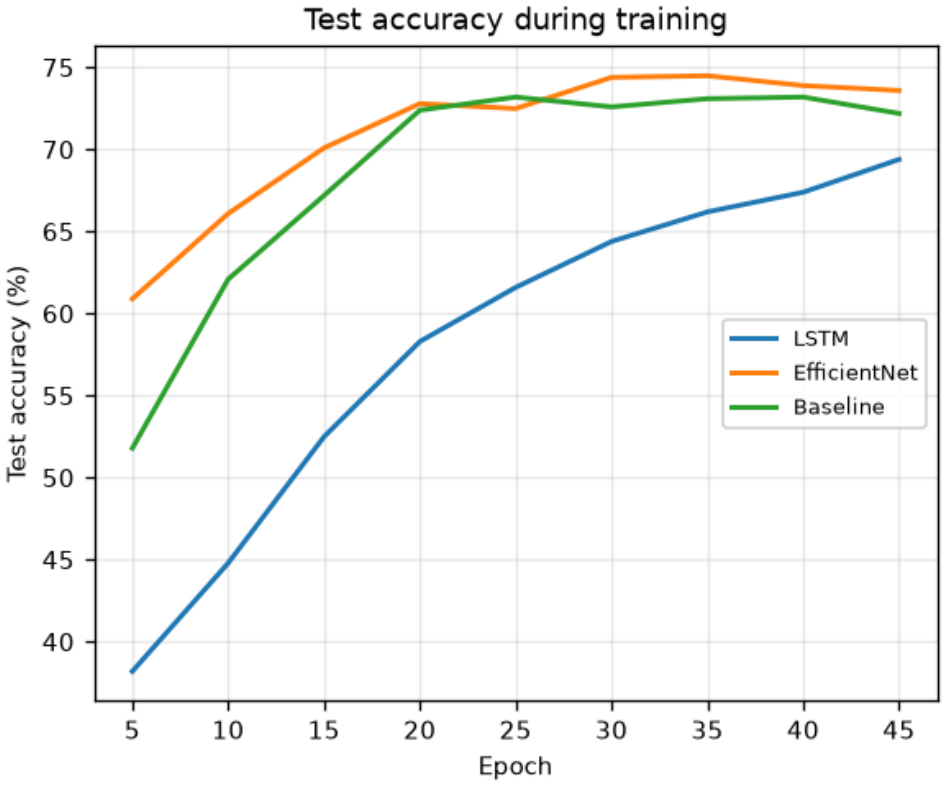


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, Baseline attains a test accuracy of 58.1% at epoch 10. EfficientNet reaches 40.9% test accuracy at epoch 5. Baseline reaches 73.0% test accuracy at epoch 45.

3 Discussion

SVM reaches 90.6% accuracy on Tiny-ImageNet. The differences between the models are discussed in the next section.